Module 4: Physics-Informed Neural Networks for SEIRD Epidemiology

Mathematical Modelling Group Project — Sierra Leone Ebola 2014–15

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1. Introduction and Motivation

Surveillance systems for Ebola report two cumulative quantities at irregular dates: confirmed cases $C(t)$ and confirmed deaths $D(t)$. Every other epidemiologically relevant quantity—the susceptible fraction S, the incubating population E, the currently infectious I, and most critically the time-varying transmission rate $\beta(t)$—is latent. Classical approaches handle this latency by assuming a parametric shape for $\beta(t)$ in advance (a step function, an exponential decay, or piecewise-constant segments). Each assumption has a qualitative cost: a step can never anticipate a second wave; an exponential can only decrease monotonically.

Physics-informed neural networks (PINNs) offer a fundamentally different strategy. Two small neural networks jointly represent the state trajectories and $\beta(t)$ continuously, and are trained to minimise a composite loss function combining fidelity to empirical observations with a residual of the SEIRD ordinary differential equations evaluated via automatic differentiation. No predetermined functional form for $\beta(t)$ is imposed. The ODE constraint propagates observational signals from cases and deaths into the unmeasured compartments: by enforcing consistency with epidemiological dynamics, the network is compelled to reconstruct latent states E and I faithfully.

2. Architecture and Inductive Biases

2.1 Observable Re-parameterisation

Let $C(t) = I(t) + R(t) + D(t)$ denote the cumulative symptomatic incidence. Differentiating yields $dC/dt = \sigma E$. The standard SEIRD equations can therefore be rewritten using (E, I, C, D):

$$dE/dt = \beta(t)(S/N)I - \sigma E, \qquad dI/dt = \sigma E - (\gamma + \mu)I, \qquad dC/dt = \sigma E, \qquad dD/dt = \mu I$$

with algebraic recoveries $R = C - I - D$ and $S = N - E - C$. Consequently, the mass conservation identity $S + E + I + R + D = N$ holds identically across all candidate network outputs, converting a difficult manifold constraint into an architectural invariant.

2.2 Dual-Network Architecture

The framework deploys two decoupled feed-forward networks:

- **State network** $g_\theta(\tau)$: 3 hidden layers of 48 tanh units with 4 outputs parameterising normalised time $\tau \in [0, 1]$:

$$e = e_0\, e^{\tau g_1}, \qquad i = i_0\, e^{\tau g_2}, \qquad c = c_0\, e^{\tau g_3}, \qquad d = d_0 + \tau\, g_4$$

- **Transmission network** $h_\varphi(\tau)$: 3 hidden layers of 32 tanh units with $\beta(\tau) = \mathrm{Softplus}(h_\varphi(\tau) + b) > 0$.

2.3 Inductive Biases

Six foundational inductive biases ensure physical validity prior to parameter optimisation: (1) *Hard initial conditions* via exponential factors ensure zero gradient error at τ=0; (2) *Strict positivity* eliminates biologically impossible negative populations; (3) *CFR parameterisation* $\mu = \gamma\, \mathrm{CFR} / (1 - \mathrm{CFR})$ with $\mathrm{CFR} \in (0.2, 0.8)$ blocks the degenerate $\mu \to 0$ branch; (4) *Fourier feature embeddings* resolve spectral bias; (5) *Relative ODE residuals* prevent early-stage dynamic attenuation; (6) *Smoothness regularisation* stabilises derivative ill-conditioning.

3. Loss Function and Training Protocol

3.1 Composite Loss Formulation

The multi-objective loss function balances data fidelity, differential dynamics, and functional regularity:

$$L = L_C + L_D + L_{ode} + w_{sm} L_{smooth}$$

where empirical terms are computed over discrete surveillance dates τ_k:

$$L_C = (1/n_C) \Sigma [c(\tau_k) - C_k/K]^2, \quad L_D = (1/n_D) \Sigma [d(\tau_k) - D_k/K]^2$$

The physics loss is evaluated over $N_f = 200$ interior collocation points using reverse-mode automatic differentiation:

$$L_{ode} = (1/N_f) \Sigma [((e' - Tf_e)/e)^2 + ((i' - Tf_i)/i)^2 + ((c' - Tf_c)/c)^2 + ((d' - Tf_d)/d)^2]$$

$$L_{smooth} = (1/N_\beta) \Sigma (d\beta/d\tau)^2$$

Here K represents the normalization scale (peak cumulative cases in training). Hyperparameters are fixed at $w_{data} = w_{ode} = 1$, with smoothness weight $w_{sm} = 10^{-4}$ for synthetic experiments and 10^{-3} for real outbreak data.

> **Remark on Relative vs. Absolute Residuals:** In epidemic outbreaks, caseloads expand across orders of magnitude. Under canonical absolute residuals $(x' - Tf)^2$, the early phase where (e, i, c) are small contributes practically zero loss gradient. The network fits only the peak, resulting in massive early dynamical drift. Normalising by state magnitude yields scale-invariant loss contributions across the entire trajectory.

3.2 Two-Stage Optimization Protocol

Optimization follows a disciplined two-stage sequence implemented entirely in IEEE float64 precision:

1. **Stage 1 — Stochastic Exploration (Adam):** Initial learning rate of 2×10^{-3} with cosine annealing over 2,000 iterations. Resampling interior collocation points each iteration injects stochastic noise that escapes flat plateaus and aligns coarse trajectory envelopes.
2. **Stage 2 — Deterministic Refinement (L-BFGS):** Quasi-Newton optimization using a strong-Wolfe line search over 2,500–6,000 iterations evaluated on a fixed linspace grid. The second-order curvature approximation resolves stiff ODE couplings and converges to tight gradient tolerances.

Point forecasts are constructed from an ensemble of 5 independent seeds. For out-of-sample forecasting, β is held constant at its trailing 14-day mean, combined with the learned CFR parameter μ, and propagated via DOP853.

4. Synthetic Experiments and Results

4.1 Experiment E2: Forward Solution Verification

With baseline parameters $\beta = 0.4858$ day^{-1} ($R_0 = 2.0$), $\mu = 0.1000$ day^{-1}, and initial fractions (0.99, 0.005, 0.005, 0, 0), the PINN was trained purely on physics loss L_{ode} in the absence of observation data. Across 3 independent seeds, mean relative L_2 errors against high-order DOP853 integration were 3.7×10^{-4} (s), 5.4×10^{-4} (e), 5.1×10^{-4} (i), 1.1×10^{-3} (r), and 7.9×10^{-4} (d). The maximum pointwise error across all compartments remained under 2.0×10^{-3}, with conservation maintained at 2×10^{-16}.

4.2 Experiments E3 and E4: Inverse Recovery from Noisy Reports

A synthetic epidemic incorporating an intervention step reducing β from 0.55 to 0.25 at day 30 was observed through noisy cumulative counts (1% and 5% multiplicative noise). The PINN reconstructed $\beta(t)$, μ, initial states, and unobserved trajectories $s(t)$, $e(t)$, $i(t)$.

Table 1: E3/E4 Inversion metrics from noisy cumulative cases and deaths (mean ± SD, 5 seeds).

Target Metric	1% Multiplicative Noise	5% Multiplicative Noise
Transmission $\beta(t)$ relative L_2	11.0% ± 0.7%	11.6% ± 0.6%
$\beta(t)$ RMSE outside transition (>5d)	0.022 ± 0.005 day^{-1}	0.025 ± 0.005 day^{-1}
Mortality rate μ relative error	4.6% ± 0.3%	5.4% ± 1.0%
Latent Susceptible $s(t)$ relative L_2	0.36% ± 0.03%	0.46% ± 0.09%
Latent Exposed $e(t)$ relative L_2	5.8% ± 0.6%	6.4% ± 0.7%
Latent Infectious $i(t)$ relative L_2	5.8% ± 0.4%	6.5% ± 0.8%

4.3 Ablation Analysis (Experiment E6)

Table 2: Systematic architectural ablations under 5% noise (mean of 2 seeds).

Configuration	$\beta(t)$ RMSE	Hidden $e(t)$ rel. L_2	μ rel. error
Full Architecture	**0.042**	**0.070**	**0.054**
Overweighted physics ($w_{ode} = 10$)	0.123	0.207	0.143
Adam only (8000 iters, no L-BFGS)	0.243	0.957	0.176
Ablate Fourier features	0.058	0.083	0.045

5. Conclusion and Critical Discussion

The SEIRD-PINN successfully resolves the epidemiological observability paradox: from noisy, aggregated reports of cumulative cases and deaths alone, it recovers continuous non-monotonic transmission trajectories $\beta(t)$, the case-fatality parameter μ, and latent infection compartments. The ablation confirms that second-order L-BFGS refinement and relative residual scaling are mathematically essential for convergence.

References

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